CS532 Quiz 1: RANSAC and Two-view Geometry

Name: _____

Gambling. A parlay is a single bet that combines two or more individual wagers, known as "legs," into one ticket. To win a parlay, every single leg must be successful. While this offers the potential for a much larger payout than a single bet, the risk is also significantly higher, as a loss on even one leg results in losing the entire parlay.

- **Q1:** What are the odds of winning a 3 coin-flip parlay assuming a fair coin?

  Solution:

  Each leg has $\frac{1}{2}$ probability of success. So to win a parley with 3 coin flips, we get

  $\mathbf{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}}$

- **Q2:** Concerned about a potential gambling addiction you limit yourself to playing one parlay a day. What are the odds of you winning at least one 3 coin-flip parlay in a week?

  Solution:

  Each 3-coin parlay has $\frac{1}{8}$ chance to win, so $\frac{7}{8}$ chance to lose.

  So the chance of never winning a 3-parlay coin flip for a week is $(\frac{7}{8})^7$.

  So the probability of winning at least once a week is $\mathbf{1 - (\frac{7}{8})^7}$

3D Object Tracking. Assume you have a pair of static cameras tracking an object in 3D. That is, each camera can estimate the 3D pose of the object $[\mathbf{R}_i, \mathbf{t}_i]$ with respect to its own camera reference system.

- **Q3:** Is it possible to estimate the relative pose of the cameras from a pair of concurrent object pose estimates (one per each camera)? Justify your answer.

  Solution:

  It is possible to estimate the relative pose of the cameras from a pair of concurrent object pose estimates.

  Let the object (O) to camera 1 (C_1) transform be $T_{C_1}^{O}$.

  Let the object (O) to camera 2 (C_2) transform be $T_{C_2}^{O}$.

  We can just stack these two transforms together to get $T_{C_1}^{C_2}$:

  $$\mathbf{T}_{\mathbf{C_1}}^{\mathbf{C_2}} = \mathbf{T}_{\mathbf{C_1}}^{\mathbf{O}}(\mathbf{T}_{\mathbf{C_2}}^{\mathbf{O}})^{-\mathbf{1}} = \mathbf{T}_{\mathbf{C_1}}^{\mathbf{O}}\mathbf{T}_{\mathbf{O}}^{\mathbf{C_2}}$$

- **Q4:** Is it possible to estimate the 3D plane formed by the object and the two observing cameras? Justify your answer.

  Solution:

  Yes, it is possible to estimate this plane:

  Let's assume the Cam 1 to be at world origin. Then the object O in cam 1 coordinate system is just t_1

  We also have the relative transformation between the 2 cameras from the previous question. This means that C_2 is just the translation part of the transform $T_{C_1}^{C_2}$, and $C_1 = (0, 0, 0)$:

  $$\begin{aligned} C_1 &= (0, 0, 0) \\ C_2 &= t_{C_1}^{C_2} \\ O &= t_1 \end{aligned}$$

  We can get the normal of these three coplanar points from the cross products, as $n = (O - C_1) \times (C_2 - C_1)$.

  From this, we get the plane equation as $\mathbf{n}^\intercal\mathbf{x} + \mathbf{d} = \mathbf{0}$